

DAILY DRILLING INCIDENT REPORT — WELL NHK-118 (WELL-007)

Well ID:	WELL-007 (Offset Well NHK-118)
Operator:	Oil India Limited (Upper Assam Field)
Formation Target:	Barail Main Formation
Incident Depth:	2,920.0 m TVD
Report Type:	Daily Drilling Report (DDR)
Date of Event:	14-Nov-2024

1. Executive Incident Summary

During wiper trip at 2920m in Barail Main Formation, drillstring experienced sudden torque elevation from 14 kN.m to 42 kN.m followed by complete mechanical rotation freeze. Reactive shale sloughing and cutting accumulation observed in tight hole section.

2. Physical Telemetry Parameters

Rate of Penetration: 8.5 m/hr | WOB: 32 klbs | Rotary Speed: 45 RPM (decreased from 110 RPM) | Standpipe Pressure: 2,450 psi | Mud Weight: 1.26 sg WBM.

3. Root Cause Analysis

Severe shale sloughing and hole pack-off in Barail formation interval due to prolonged exposure to water-based mud. Depleted sand lenses created tight hole differential pressure conditions.

4. Applied Mitigation Procedure

1. Immediately spotted 50 bbl high-lubricity pipe-freeing pill (oil-based surfactant blend) across reactive Barail section.
2. Increased mud pump circulation rate from 400 gpm to 530 gpm to flush cuttings.
3. Worked drillstring with downward jarring using 115 tons maximum overpull while maintaining slow string torque.

5. Final Operational Outcome

Drillstring was successfully freed after 26 hours of soaking pill and jarring without requiring string back-off or sidetrack. Hole back-reamed and conditioned prior to casing run.